Large-scale Frontends with Mixed Reality

Simon Scheidegger CSL Behring

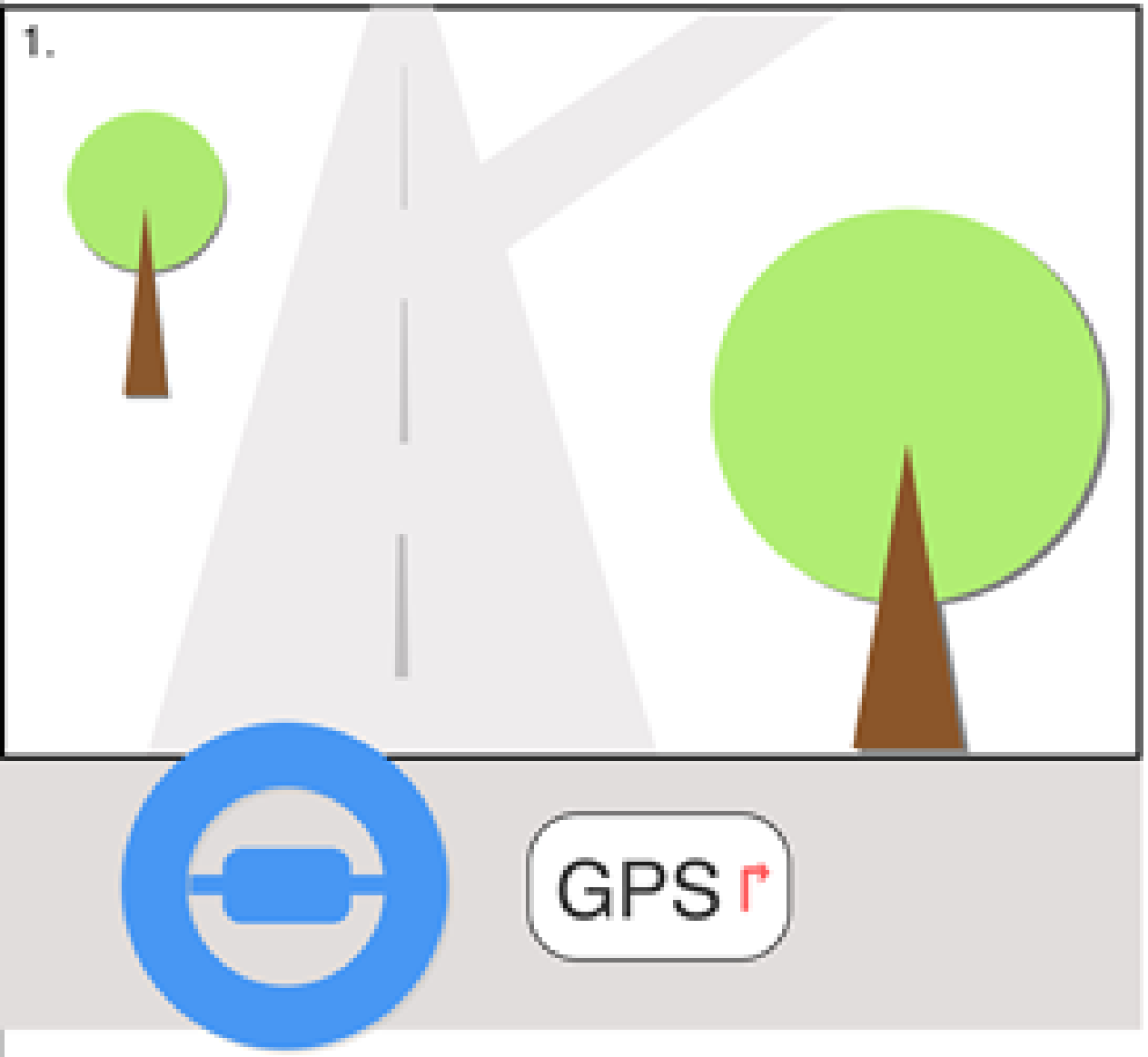
Paul Affentranger afca.

6:18
Saturday, June 18

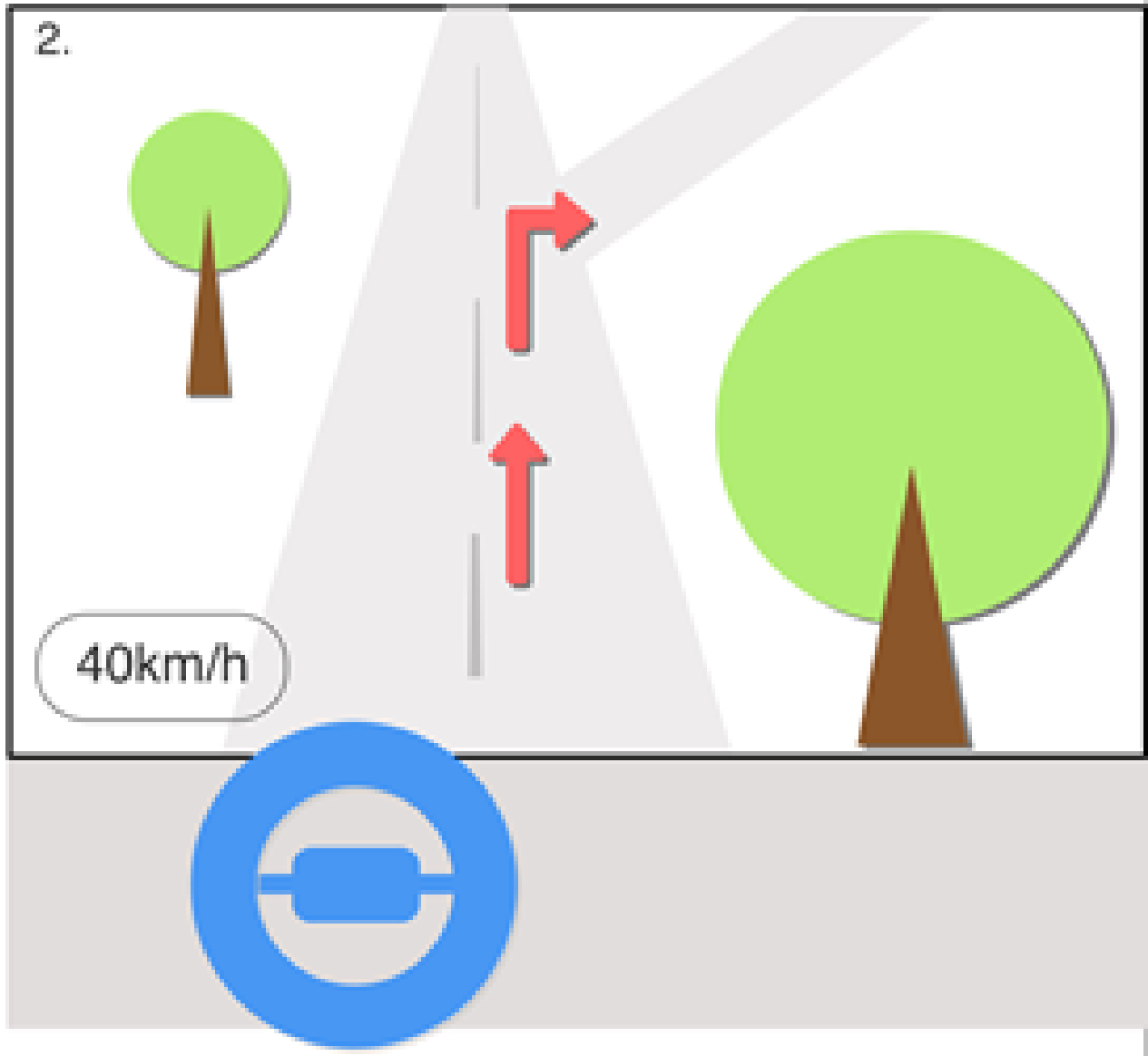
SONY
10:35

Firstline Workers

- Continue to be overlooked
- 48% of companies report that the majority of their firstline workers are not digitally connected
- UI often not adapted



Higher Cognitive Load

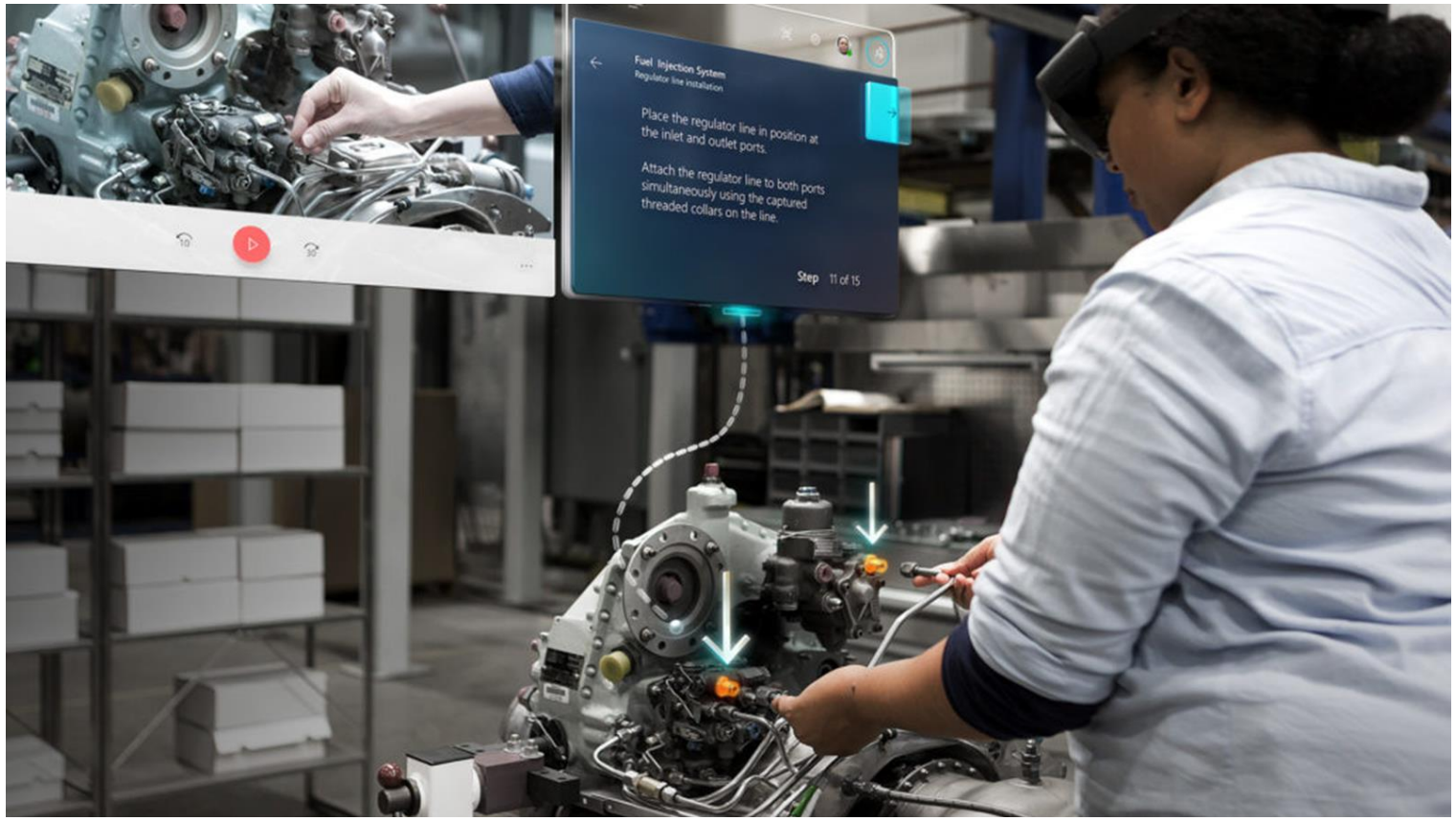


Lower Cognitive Load

Source: collectar.io

Low Cognitive Load

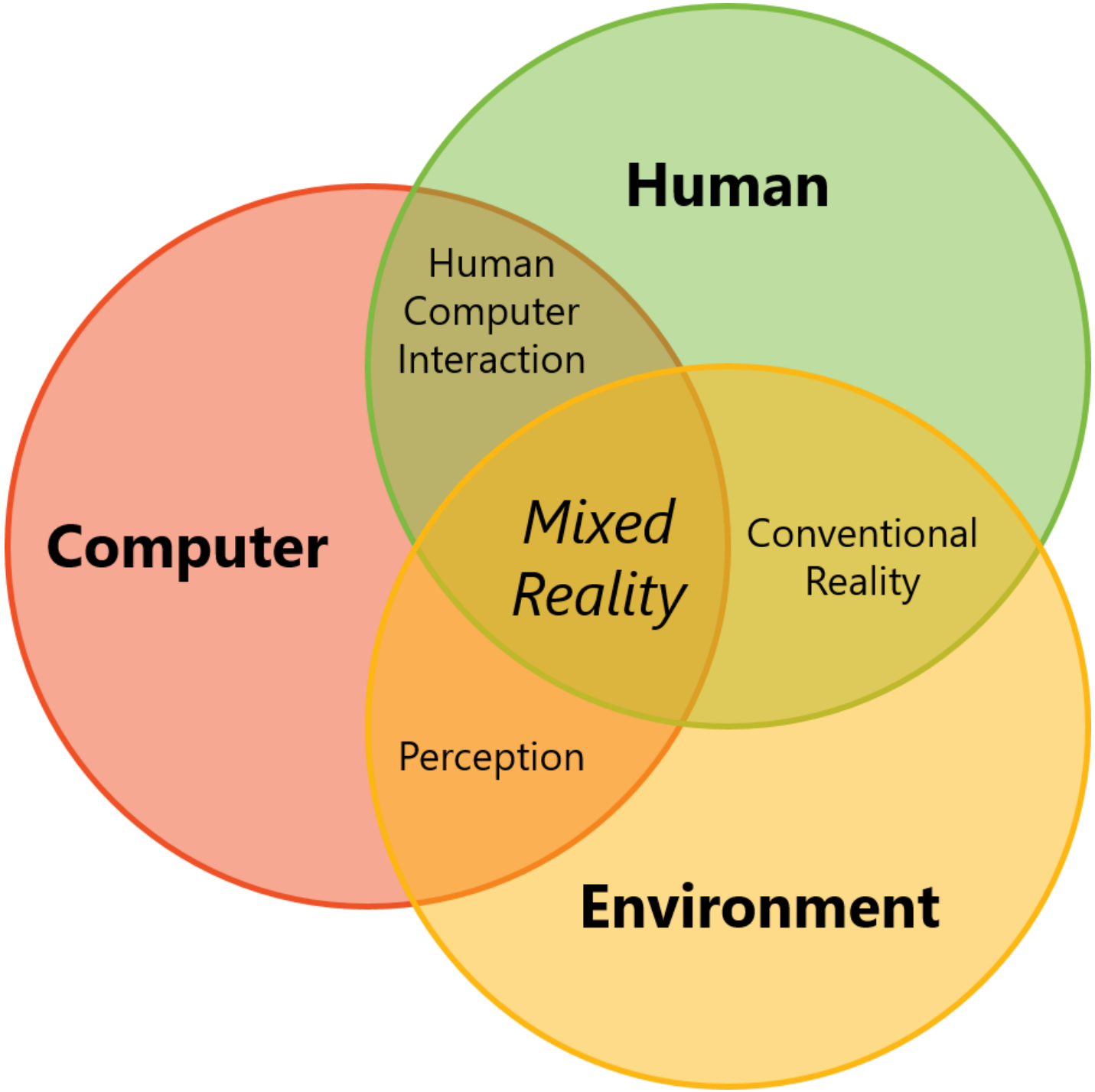
Hands free

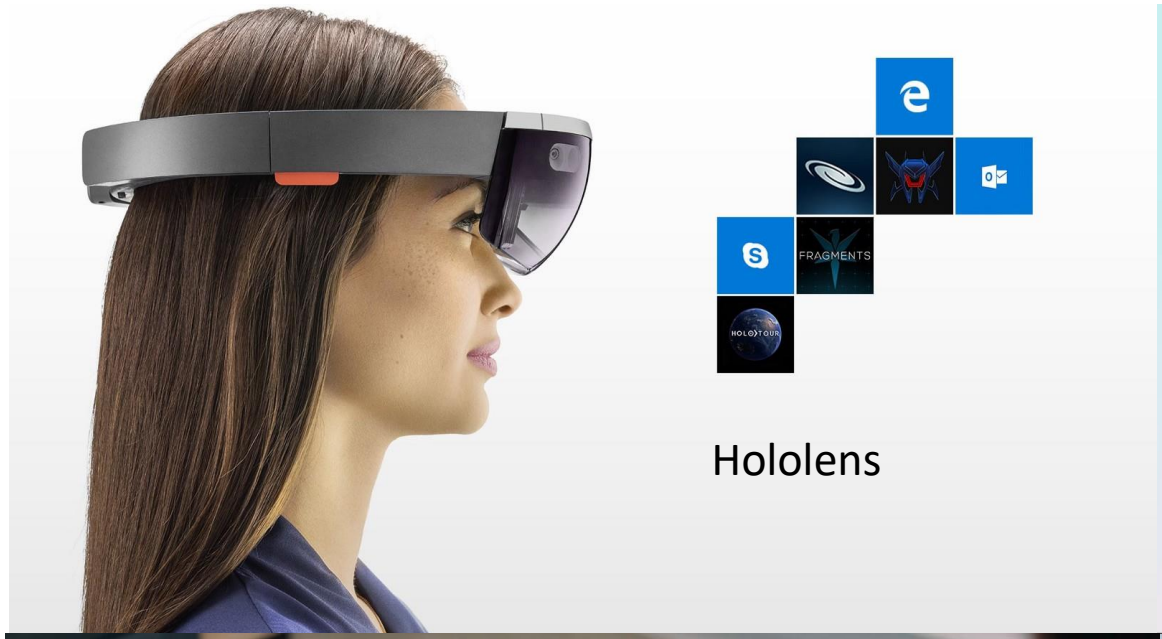

Fuel Injection System
Regulator line installation
Place the regulator line in position at the inlet and outlet ports.
Attach the regulator line to both ports simultaneously using the captured threaded collars on the line.
Step 11 of 15

VR

AR

MR

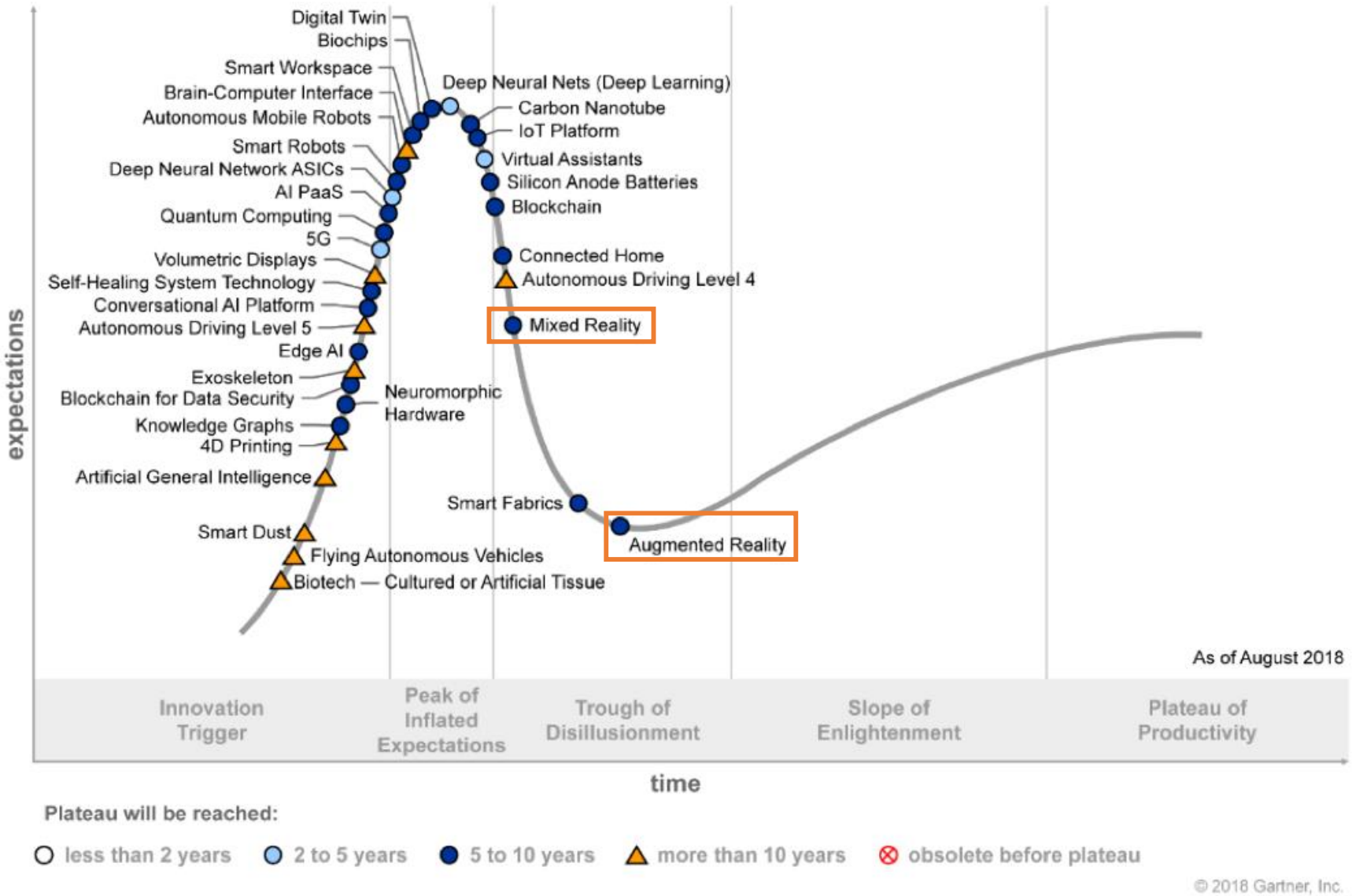

Human
Computer
Environment
Human Computer Interaction
Mixed Reality
Conventional Reality
Perception



Hololens

Apple
ARKit

Google
ARCore



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30+ Countries
Of operations around the world

US$ **6.9+** Billion
In annual revenue

8
Manufacturing sites

Australia
China
Germany
Switzerland
United Kingdom
United States

19,000+
Employees around the world

US$ **2.6** Billion
In R&D investments in last 5 years advances exciting pipeline

1,400+
R&D employees

170+
Plasma collection centres across Europe and North America

CSL Behring Bern

Privigen

Hizentra

Albumin

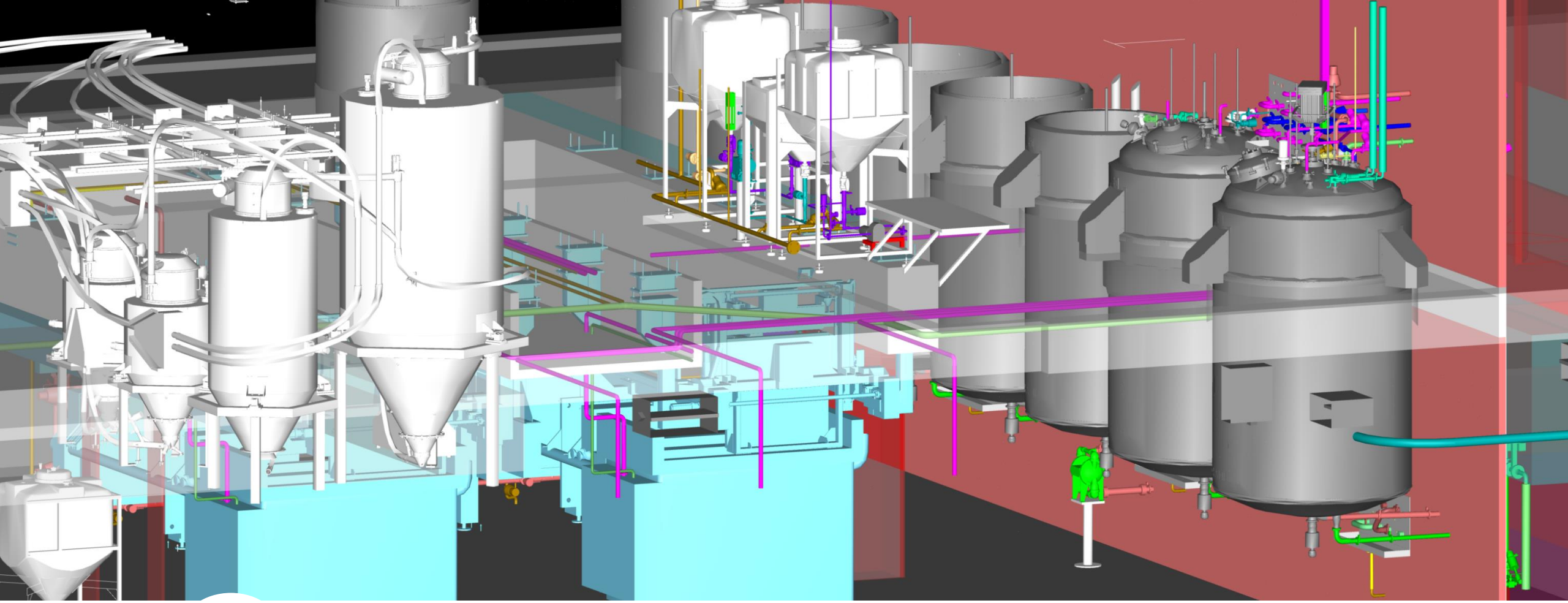
Rhophylac

Die Arbeit in der Produktionszone

- Mehrstufige Prozesse
- Komplexe Aufgaben für Mitarbeiter
- Hoher Leistungsdruck
- Strenge Vorgaben und validierte Prozesse / MES

Schwierigkeiten

- Fokus der Systeme auf korrekten Ablauf, nicht Benutzerfreundlichkeit
- Übersicht oft schwierig. MES unterstützt nicht bei Planung
- Änderungen erfolgen im Validierten/registrierten System langsam

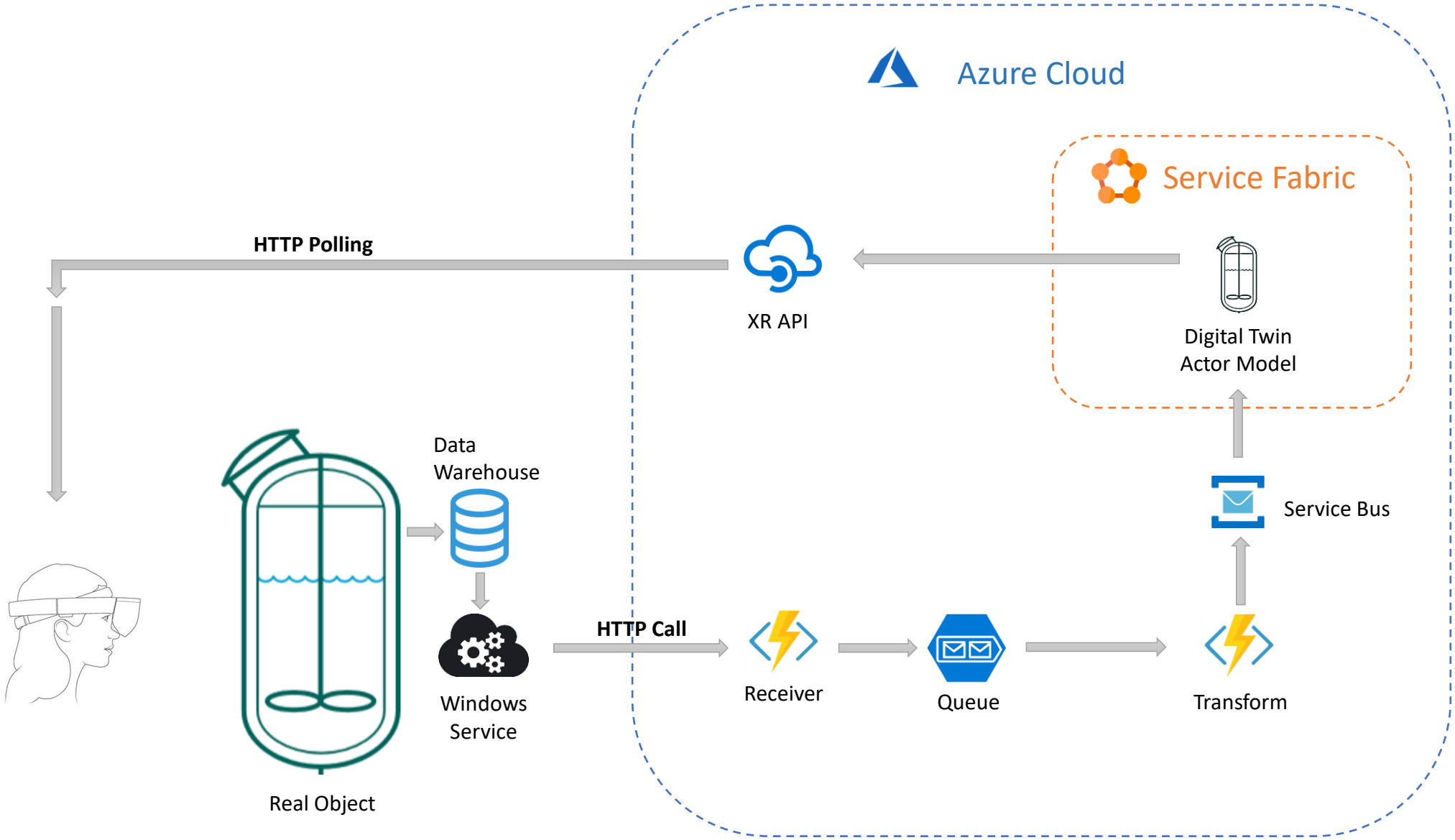


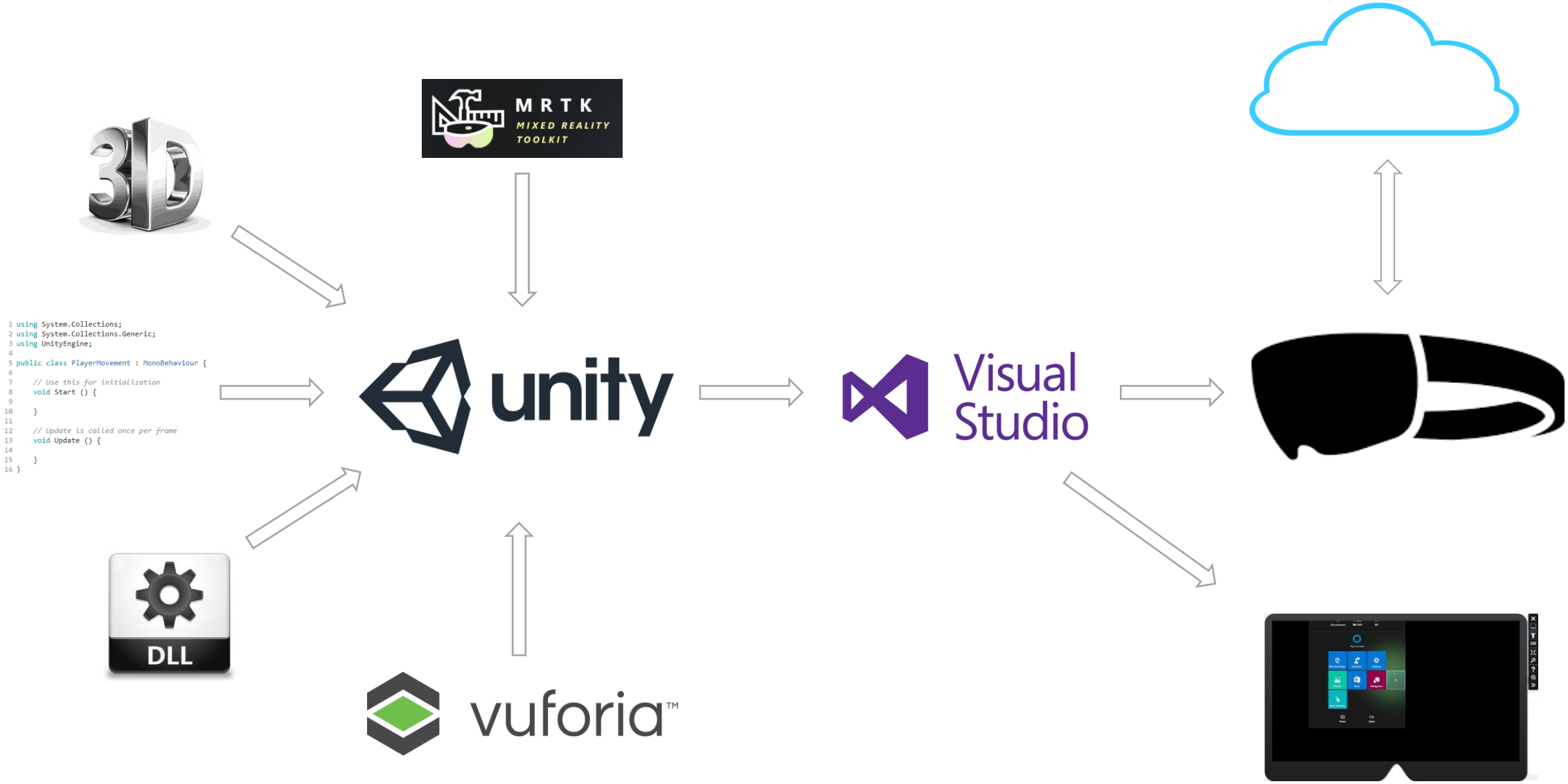
3D Model for Spatial Information

1U 007
LAB 1U 130
M. Air Lock
1U014
LAB 009
LAB 1U 013
LAB 1U 010
CIP1
CIP2
CIP3
CIP4
RGW
FP5
Waschk.
ZWG 1U 311
Technical Area
ZWG 1U 315

FT 16
FT 14

Architektur




MRTK
MIXED REALITY
TOOLKIT
3D
using System.Collections;
using System.Collections.Generic;
using UnityEngine;
public class PlayerMovement : MonoBehaviour {
// Use this for initialization
void Start () {
}
// Update is called once per frame
void Update () {
}
}
unity
Visual
Studio
DLL
vuforia™

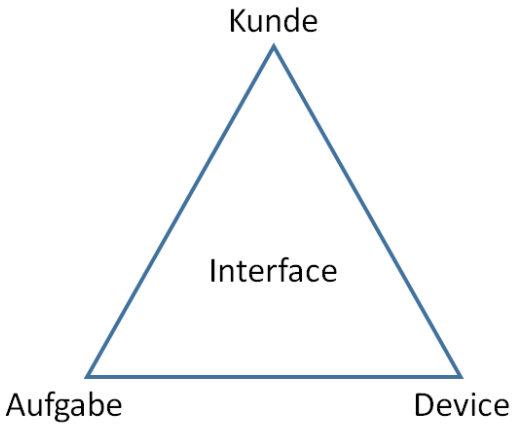
PlasmiX

CSL Behring

2018 • Built by afca.

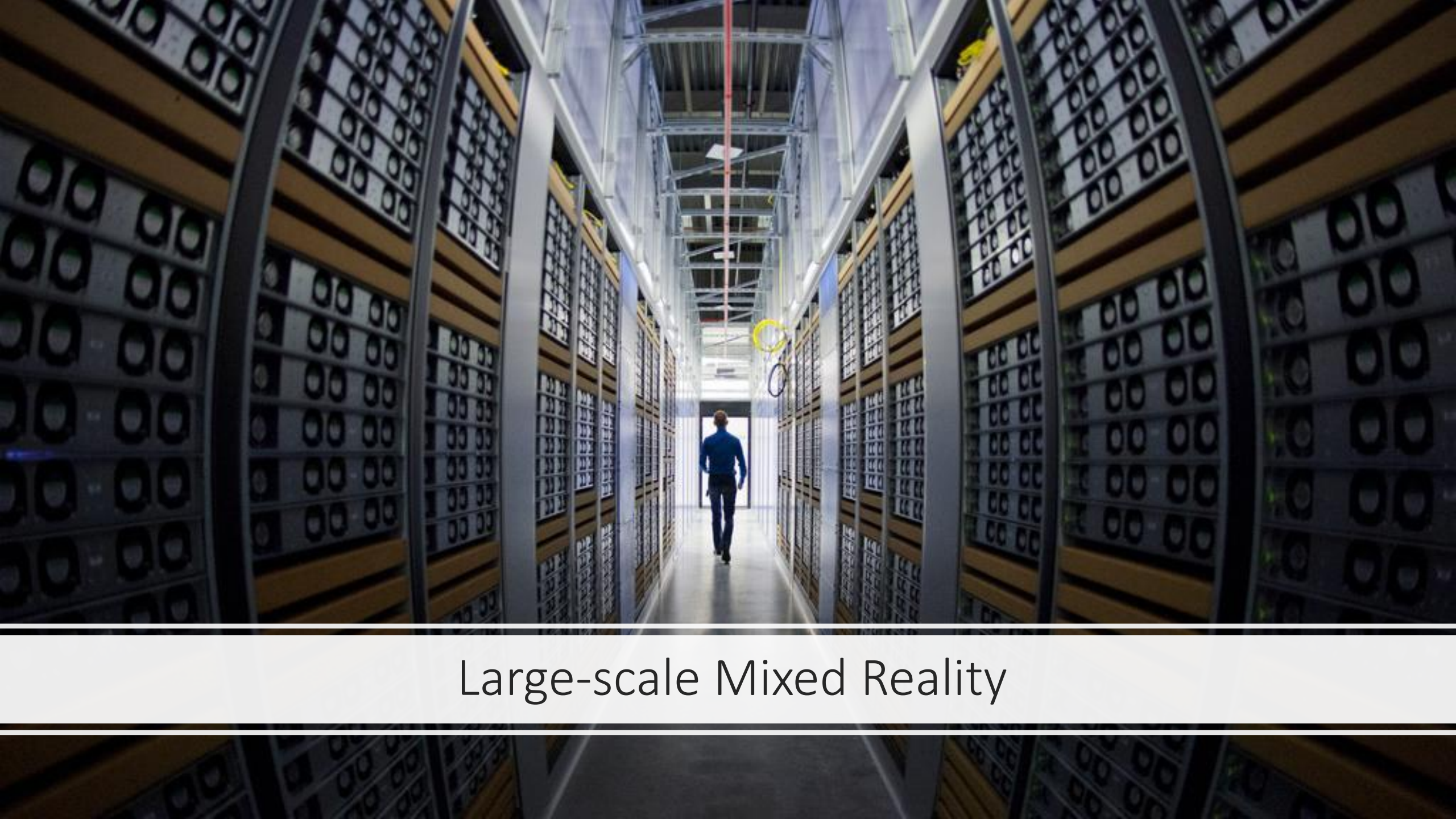
FT14
P100008487 LOT
-01:04
A
CIP Tank
IV
Fällung
C
CIP vor Filtration
Filtration
Reinigung COP
Mischzeit 210min
03:56 Warten auf
pH Kontrolle nach EtOH
03:56 Warten auf
Kontrolle nach EtOH
kg
°C
0.00 pH
300min

Experience


Kunde
Interface
Aufgabe
Device

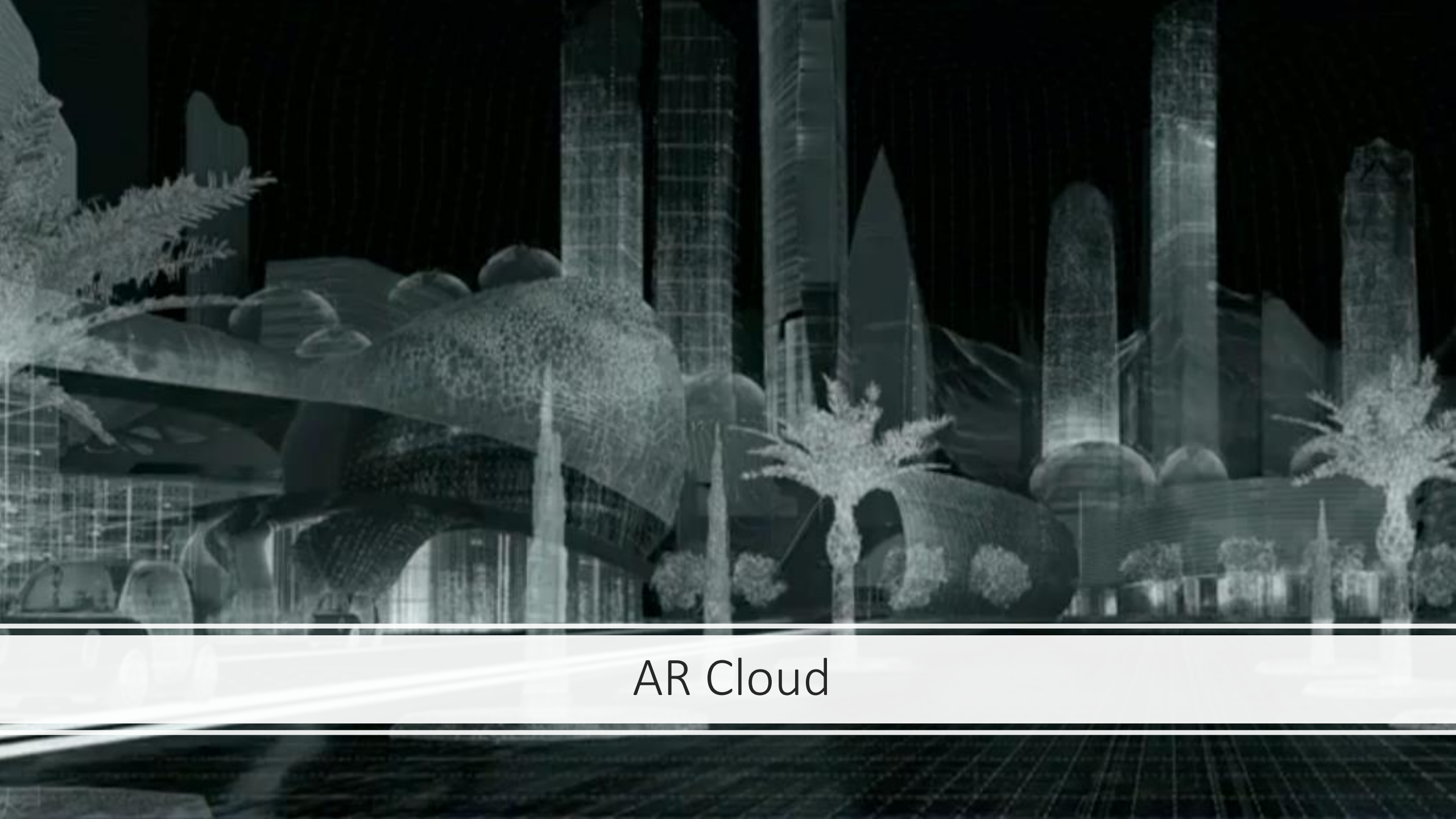
Challenges

- Device
  - Heavy
  - 2-3h battery live
  - Rrestricted field of view
- Data
  - Record based instead of Events
  - High latency
- 3D Data (Environment)
  - To many details



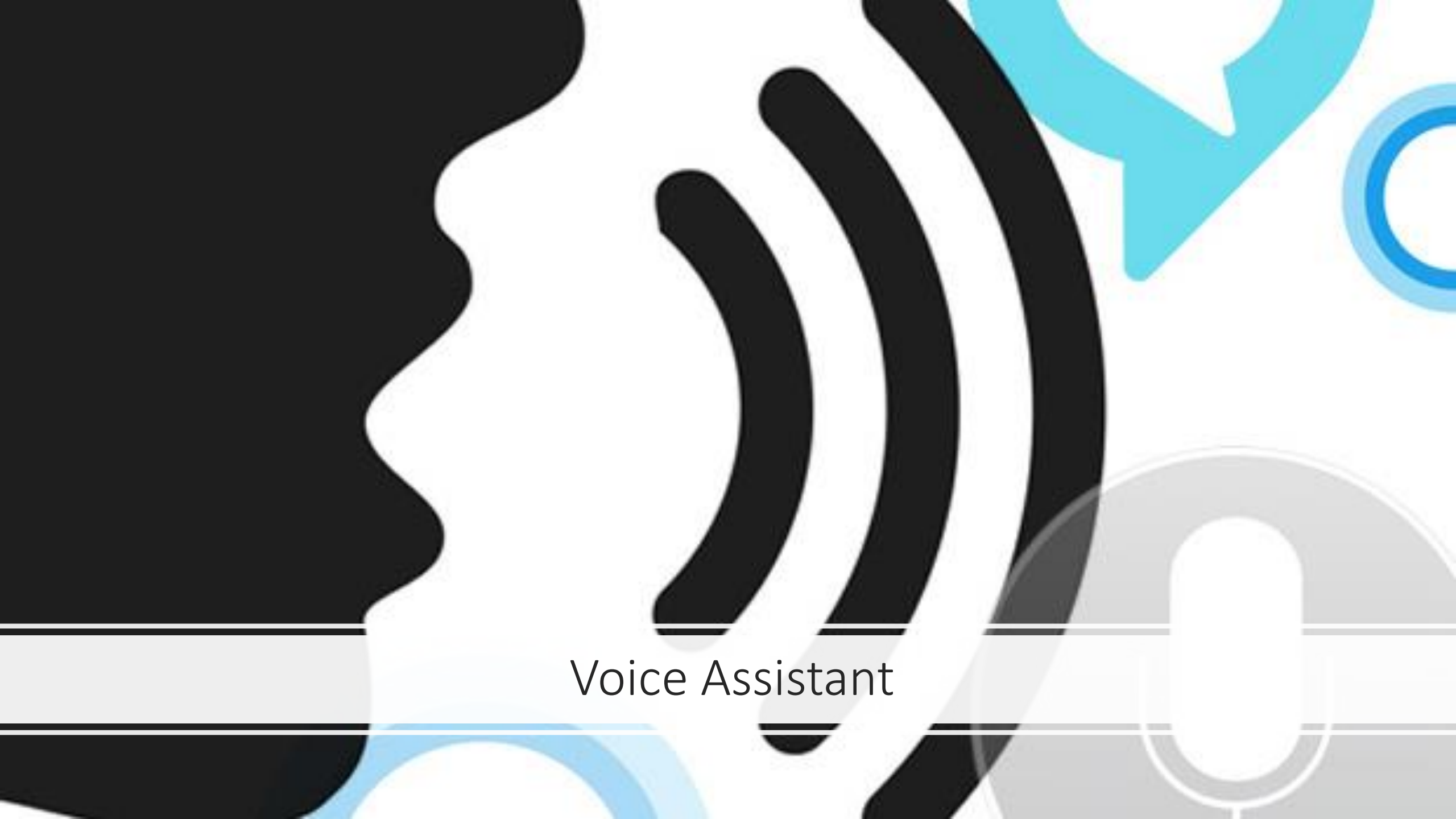
Large-scale Mixed Reality

Break down silos



AR Cloud

IoT + AI



Voice Assistant

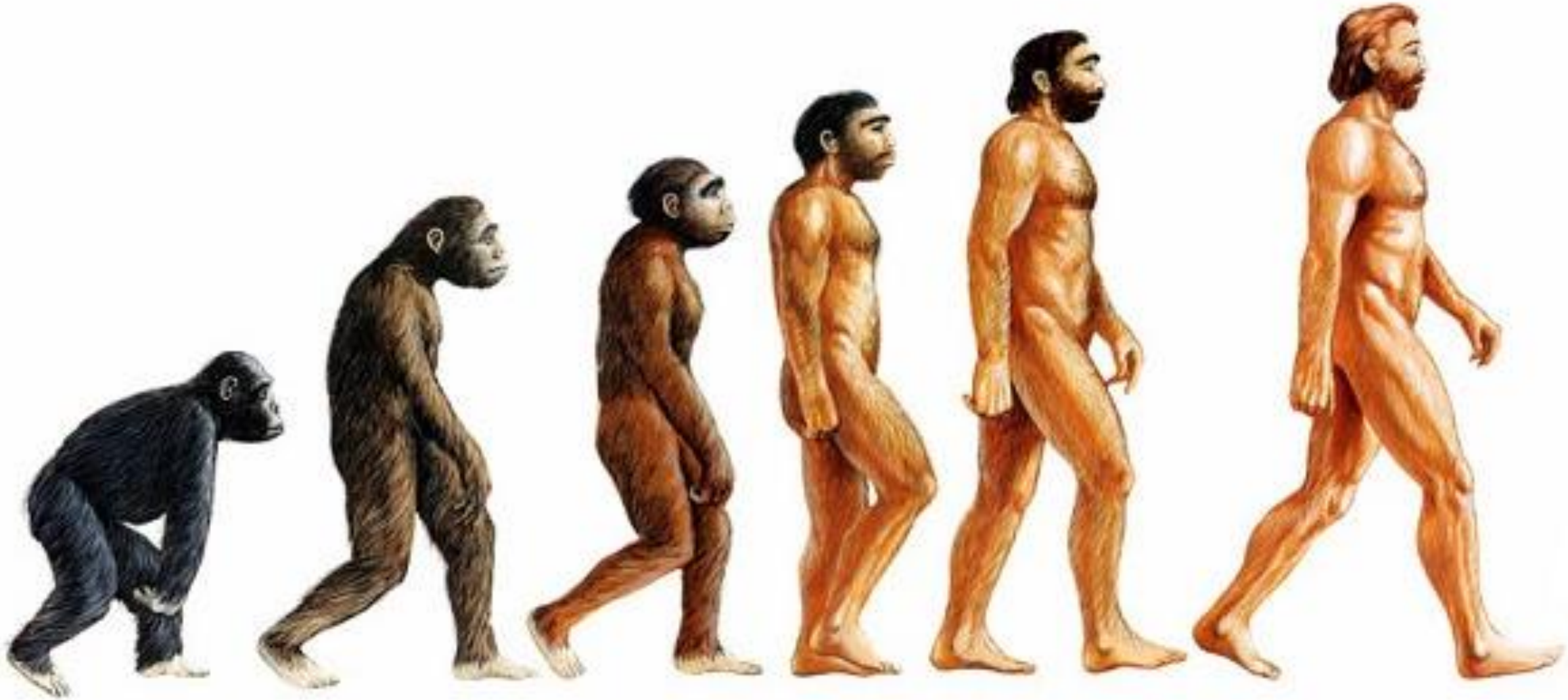
Context is important

How to start

- «Solve» chicken – egg problem
- Think big - start small
- Concentrate on real value
- Start with Frontline Workers
- Start with things you can’t do with 2D UI

NE
PARADIGM
AHEAD

Human Nature is key



Thank you!

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afca.